Supplement: N-level hierarchical POMDP methods

Supplement: N-level hierarchical POMDP methods I

This supplement specifies the generic N-level architecture that sec. 13.11 exercises at depth three: how the meta-context (L3), context (L2), and location (L1) factors are chained through conditioned priors, what the declarative `LayerSpec` interface fixes versus leaves free, and the top-down/bottom-up passes the inference runs. It answers *what the executed 3-level result is a special case of* — the reason the same code runs at other depths without new mathematics — while recording that only the declared 3-level configuration is empirically evaluated here.

The 3-level POMDP implemented in `fedference.pomdp.build_3level_world` extends the 2-level construction (sec. 13.10) by adding a top-level meta-context factor:

Generative model for an N-level hierarchy I

- **L3 (meta-context)** — 2 states (`low_threat` / `high_threat`) with initial uniform prior, gating the L2 context prior;
- **L2 (context)** — 2 states (`quiet` / `alert`) with context-conditioned L1 location priors, gating the L1 prior;
- **L1 (location)** — the standard 3x3 grid with 9 states and sensor acuity 0.85.

The conditioned priors are (see eq. 32 and eq. 33):

Generative model for an N-level hierarchy I

L3 state	L2 prior (quiet, alert)
`low_threat`	(0.50, 0.50) — uniform context
`high_threat`	(0.20, 0.80) — peaked at alert

L2 state	L1 prior
`quiet`	uniform over all 9 location states
`alert`	mass 0.60 at center cell (flat index 4), residual uniform

`fedference.pomdp.LayerSpec` and `fedference.pomdp.build_nlevel_world` implement the generic N-level version. The declarative layer specification is stored at `src/fedference/config/hierarchical_layers.yaml` and mirrors the canonical 3-level defaults (a standalone documentation artifact not read by any code path, kept in sync with the `build_3level_world` defaults). The constructor accepts depth ≥ 2; the executed empirical result in this manuscript is restricted to the declared 3-level configuration, and the leaf layer must carry `n_states == N_LOCATIONS`.

Inference algorithm across hierarchy levels I

`fedference.pomdp.nlevel_infer` performs 4 passes of top-down / bottom-up alternating minimization over all N levels:

Inference algorithm across hierarchy levels I

1. **Top-down pass** — compute the empirical prior for each level by marginalizing over the level above (eq. 32, eq. 33).
2. **L1 update** — one-step variational posterior on the observation: $q_{\mathsf{loc}} = \mathrm{softmax}(\log \widetilde{\pi}_{0,\mathrm{L1}} + \log A[\mathsf{obs},\,\cdot])$.
3. **Bottom-up pass** — update each non-leaf level's belief from the marginal evidence contributed by the level below: $\ell_j = \log(\tilde{p}^{\top}_{\mathsf{child}|\mathsf{parent}=j}\, q_{\mathsf{child}})$.

Inference algorithm across hierarchy levels I

After 4 iterations the agent broadcasts all N level beliefs; the colony federates each level independently via a log-linear pool (eq. 6).

Study parameters for the three-level run I

Study parameters for the three-level run I

Table 1: Study 7 three-level hierarchical POMDP execution parameters: agent count, seeded trial budget, observation acuity, alternating-minimization iterations, and the L3/L2/L1 state cardinalities used by the three-level condition.

Parameter	Value
Agents	4
Trials	960
Acuity	0.85
Alternating-min iterations	4
L3 meta-context states	2
L2 context states	2
L1 location states	9
Seed	0

Study parameters for the three-level run I

The executed three-level configuration is summarized in tbl. 1.